Higher National Diploma in Engineering (Mechanical/Civil)

Higher National Diploma in Building Service Engineering

First Year, First Semester Examination – 2021

CE-1102-Fluid Mechanics

Answer any five questions. **3hours**

Question 1 (Q1)

a) Derive expressions for **Total pressure force (F)** and distance for the **center of pressure (h^*)** from the free surface of a liquid for an inclined plane surface submerged in a liquid. (08 marks)

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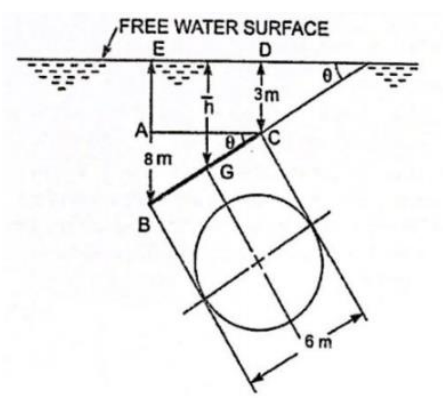
b) A **circular plate $6\ \mathrm{m}$ in diameter** is immersed in water in such a way that its greatest and least depth below the free surface are $8\ \mathrm{m}$ and $3\ \mathrm{m}$ respectively. Determine the **total pressure force** on one face of the plate and the **position of the center of pressure**. (12 marks)



Question 2 (Q2)

a) i) Explain what is **dynamic viscosity** and **kinematic viscosity** of a fluid. (03 marks) ii) Explain how does **viscosity change with temperature** of a liquid and gases with relevant equations. (03 marks)

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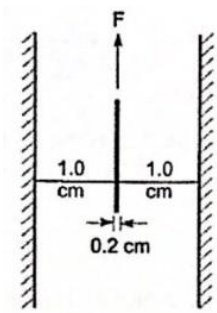
b) A vertical gap **2.2 cm wide** of infinite extent contains a fluid of viscosity $2\ \mathrm{N}\cdot\mathrm{s}/\mathrm{m}^2$ and specific gravity of 0.9. A metallic plate $1.5\ \mathrm{m} \times 1.5\ \mathrm{m} \times 0.2\ \mathrm{cm}$ is to be lifted up with a constant velocity of $0.2\ \mathrm{m}/\mathrm{s}$, through the gap, to find the force required. The weight of the plate is $50\ \mathrm{N}$. (14 marks)



Question 3 (Q3)

a) i) Briefly Explain the **continuity equation**. (03 marks) ii) Briefly Explain **Bernoulli's equation**. (03 marks)

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b) An **orifice meter** with an orifice diameter of **10 cm** is inserted in a pipe of **20 cm** diameter. The pressure gauges fitted upstream and downstream of the orifice meter give readings of $19.5\ \mathrm{N/cm^2}$ and $10\ \mathrm{N/cm^2}$ respectively. The **coefficient of discharge** for the orifice meter is given as 0.7. Find the **discharge of water** through the pipe. (14 marks)

Question 4 (Q4)

a) i) What is the **major cause of energy loss** of fluid flow in pipes? (02 marks) ii) Write **two minor causes of energy loss** of fluid flow in pipes and write equations to calculate those losses with suitable sketches. (06 marks)

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b) An oil of **specific gravity** 0.8 is flowing through a pipe of diameter **300 mm** at the rate of $600\ \mathrm{liters/s}$. Find the **head loss due to friction** and **power required** to maintain the flow for a length of $1200\ \mathrm{m}$. Take $\nu = 0.3\ \mathrm{stokes}$. (12 marks)

Question 5 (Q5)

a) Find the **drag force** exerted on a flat plate of a size $3\ \mathrm{m} \times 3\ \mathrm{m}$ when the plate is moving at a speed of $5\ \mathrm{m/s}$ normal to its plane in the water. The **coefficient of drag** is 1.2. (08 marks)

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b) Find the expression for the **power (P)**, developed by a pump when P depends upon the **head (H)**, the **discharge (Q)** and the **specific weight (w)** of the fluid, using **dimensional analysis**. (12 marks)

Question 6 (Q6)

a) A jet of water of diameter **50 mm** strikes a **fixed plate** in such a way that the angle between the plate and the jet is 30°. The **force exerted in the direction of the jet** is **1500 N**. Determine the **rate of flow of water**. (08 marks)

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b) Write short notes on the following with suitable sketches: i) **Centrifugal pump** (04 marks) ii) **Pelton turbine** (04 marks) iii) **Kaplan turbine** (04 marks)